

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Managed comprehensive **hydrogeological impact assessments** for infrastructure projects, ensuring compliance with environmental legislation and Project Ireland 2040 goals.
- Designed and deployed **groundwater monitoring programs**, collecting and analyzing critical **water quality and quantity data** to inform strategic development decisions for client projects.
- Prepared detailed **WFD, GWDTE, and Risk assessments**, contributing essential hydrogeological insights to broader **EIARs and regulatory submissions** for consultancy clients.
- Conducted site visits and technical assessments, preparing **factual and interpretative desk studies** and developing **conceptual site models** for complex environmental projects.
- Produced technical reports interpreting **hydrogeological data**, supporting the design and implementation of comprehensive site investigations for varied client needs.
- Collaborated with **clients, multidisciplinary teams**, and regulatory bodies on project scoping, planning, and programming, ensuring timely delivery and stakeholder alignment.

SKILLS

Hydrogeology & Hydrology – Hydrogeological Impact Assessments, Groundwater Monitoring, Water Quality Analysis, Water Quantity Analysis, WFD Assessments, GWDTE Assessments, Risk Assessments, Site Investigation Reports, Conceptual Site Models, Site Investigation Data Interpretation

Environmental Consulting – Environmental Legislation Compliance, EIAR Contributions, Regulatory Submissions, Project Scoping, Project Planning, Project Programming, Tender Preparation, Fee Proposals, Employer's Representative Role

Data Analysis & Reporting – Data Analysis, Technical Reporting, Interpretative Reporting, Data Visualisation

Software & Tools – Power BI, Tableau, R/RStudio, Python, SQL, Jira, Confluence

Engineering & Infrastructure – Infrastructure Planning, Water Infrastructure, Smart Cities, Rail & Urban Transit

PROJECTS

Groundwater Resource Assessment for Urban Development

- Conducted a detailed **hydrogeological impact assessment** for a proposed urban expansion project, analyzing existing groundwater levels and flow paths.
- Developed a **groundwater monitoring plan** involving the installation of 15 new piezometers and routine water sampling over a 6-month period, tracking key parameters like pH and conductivity.
- Prepared a comprehensive report detailing potential impacts on local aquifers and recommending mitigation strategies, contributing to the project's **regulatory submission**.

Water Quality Monitoring Dashboard (MSc Project)

- Engineered an ETL process using **Power Query** to consolidate water quality data from three disparate sources into a unified data model.
- Developed an interactive **Power BI dashboard** visualizing trends in chemical and physical parameters, providing insights for environmental science stakeholders.
- Implemented **DAX measures** to calculate statistical summaries and identify anomalies in water quality readings, aiding in the interpretation of collected data.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)